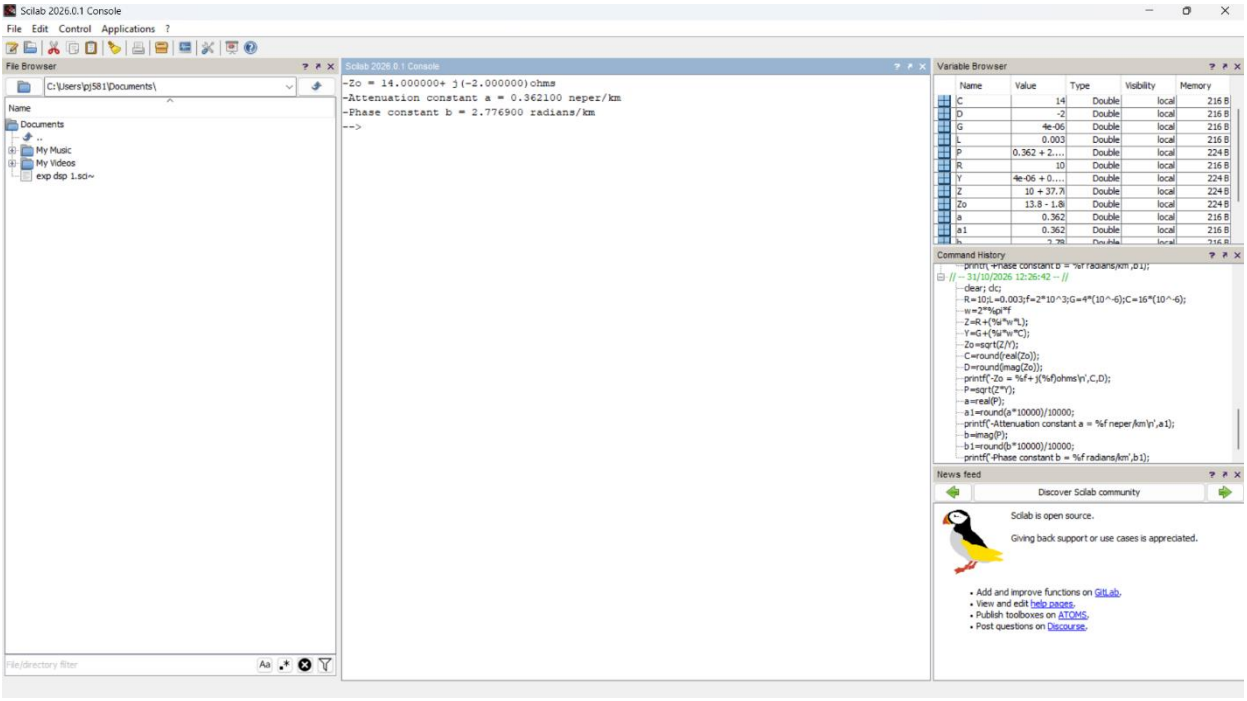
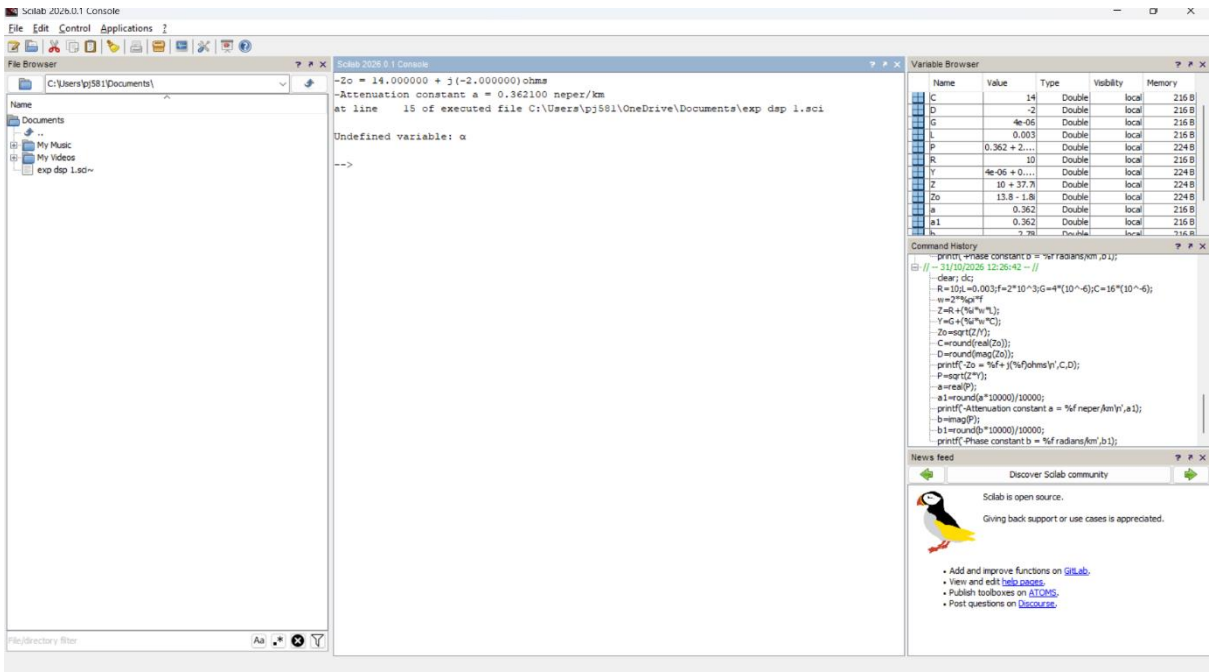
G.vinay
192312380

1.Estimation of Characteristic Impedance (Zo), and Propagation Constant (y) of aTransmission Line from Primary Constants

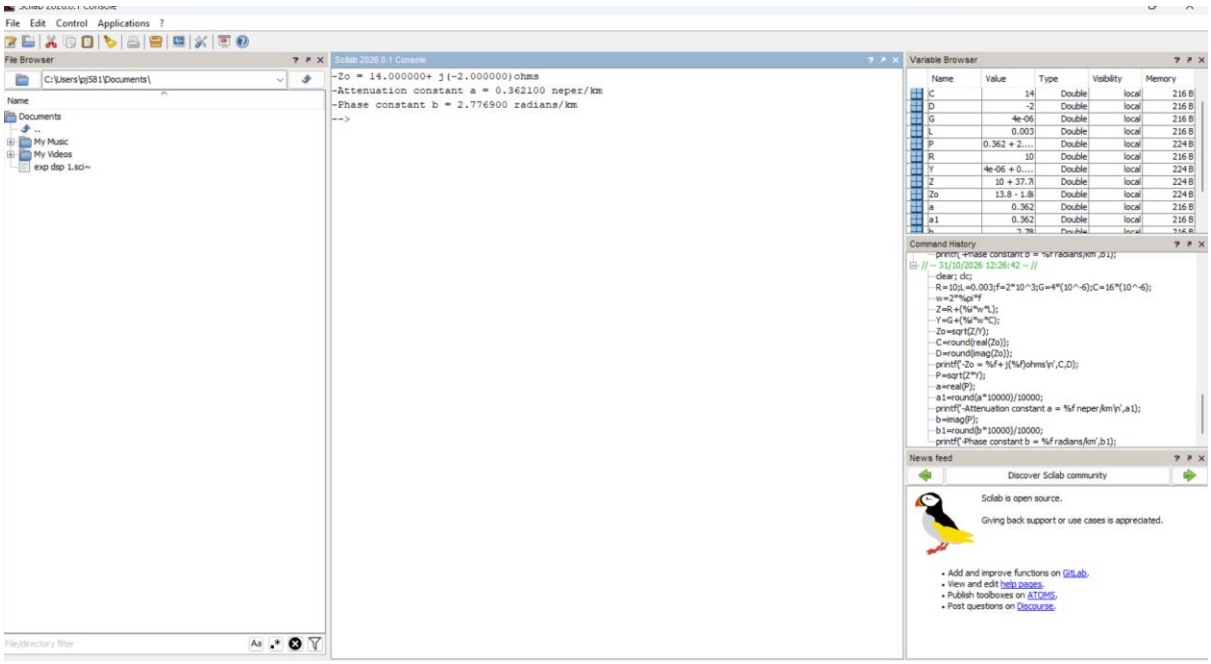
TEST CASE 1

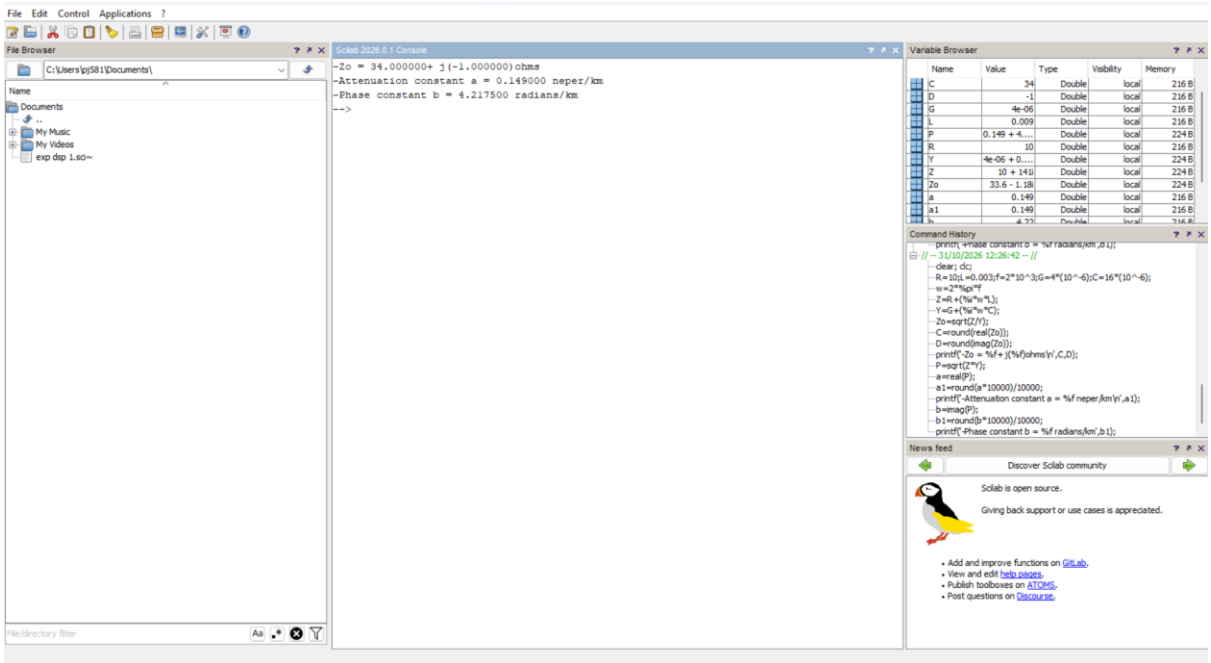


TEST CASE 2



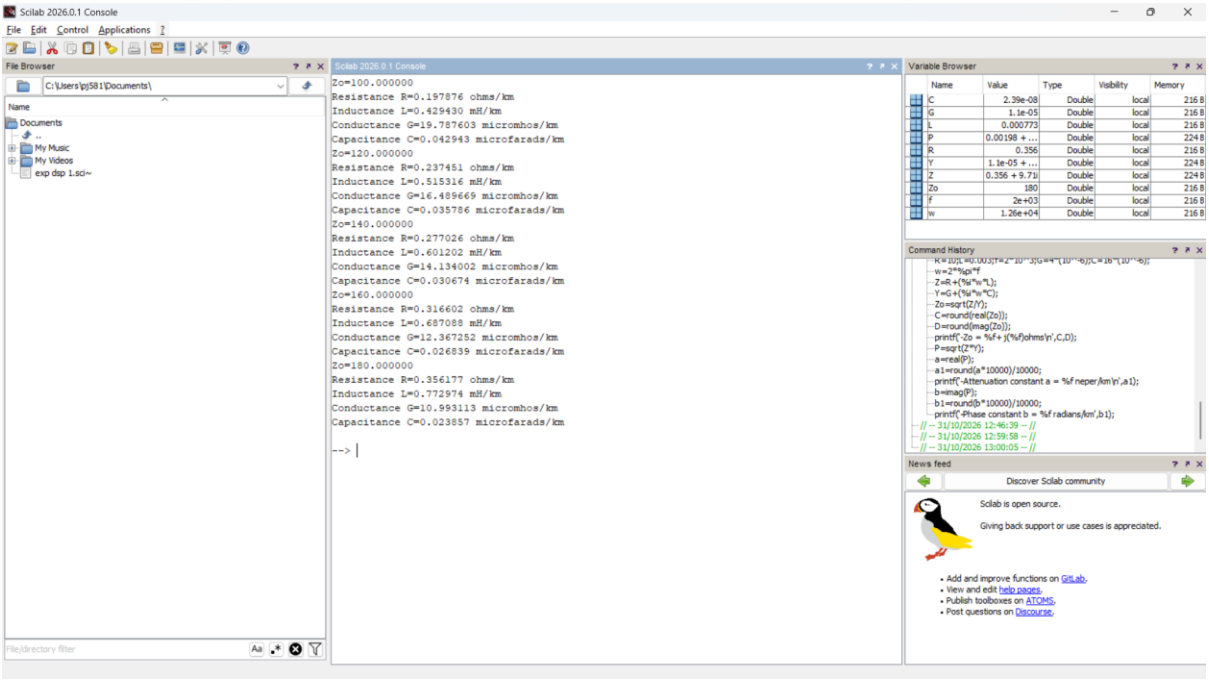
TEST CASE 3


File Edit Control Applications
File Browser
C:\Users\pj581\Documents\
Name
Documents
My Music
My Videos
exp dsp 1.sci~
Scilab 2026.0.1 Console
-Zo = 14.000000+ j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->
Variable Browser
Name Value Type Visibility Memory
C 14 Double local 216 B
D -2 Double local 216 B
G 4e-06 Double local 216 B
L 0.003 Double local 216 B
P 0.362 + 2.... Double local 224 B
R 10 Double local 216 B
Y 4e-06 + 0.... Double local 224 B
Z 10 + 37.7i Double local 224 B
Zo 13.8 - 1.8i Double local 224 B
a 0.362 Double local 216 B
a1 0.362 Double local 216 B
Command History
// -- 31/10/2026 12:26:42 -- //
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f+ j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
News feed
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Scilab is open source.
Giving back support or use cases is appreciated.
• Add and improve functions on GitLab.
• View and edit help pages.
• Publish toolboxes on ATOMS.
• Post questions on Discourse.

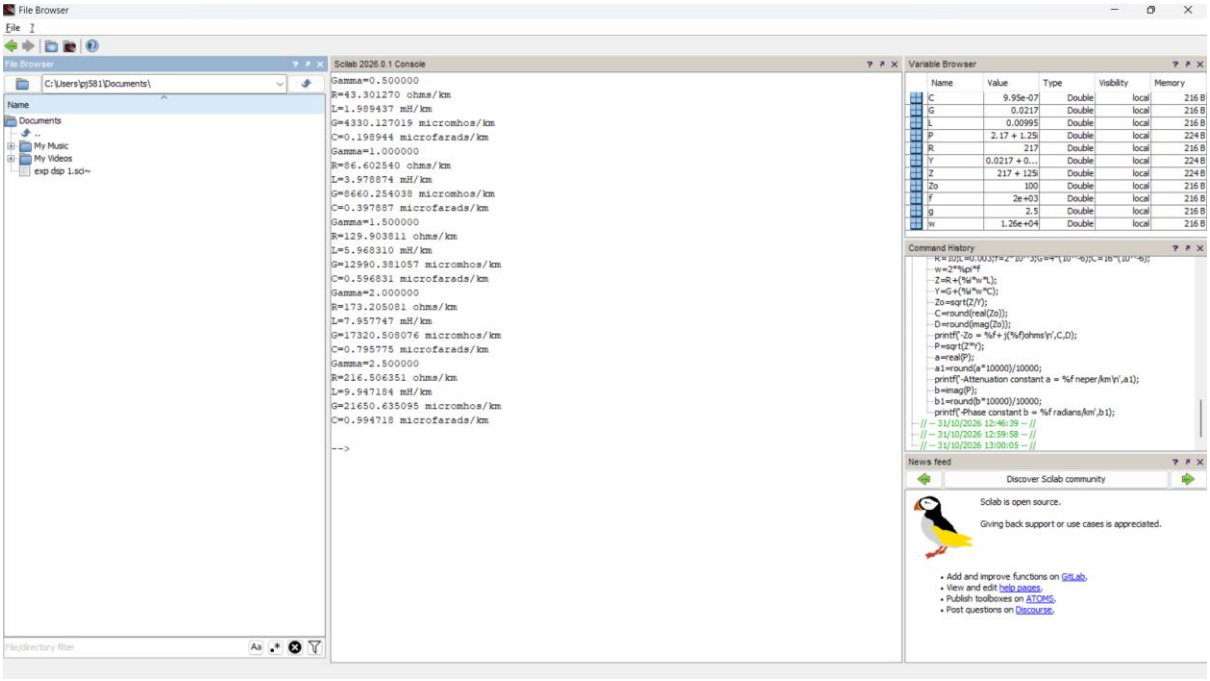

File Edit Control Applications
File Browser
C:\Users\pj581\Documents\
Name
Documents
My Music
My Videos
exp dsp 1.sci~
Scilab 2026.0.1 Console
-Zo = 34.000000+ j(-1.000000)ohms
-Attenuation constant a = 0.149000 neper/km
-Phase constant b = 4.217500 radians/km
-->
Variable Browser
Name Value Type Visibility Memory
C 34 Double local 216 B
D -1 Double local 216 B
G 4e-06 Double local 216 B
L 0.009 Double local 216 B
P 0.149 + 4.... Double local 224 B
R 10 Double local 216 B
Y 4e-06 + 0.... Double local 224 B
Z 10 + 141i Double local 224 B
Zo 33.6 - 1.1i Double local 224 B
a 0.149 Double local 216 B
a1 0.149 Double local 216 B
Command History
// -- 31/10/2026 12:26:42 -- //
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f+ j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
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Scilab is open source.
Giving back support or use cases is appreciated.
• Add and improve functions on GitLab.
• View and edit help pages.
• Publish toolboxes on ATOMS.
• Post questions on Discourse.

2.Estimation of Primary Constants (R,L,G &C) from Secondary Constants (Zo & y)of an open wiretransmission line

TEST CASE 1

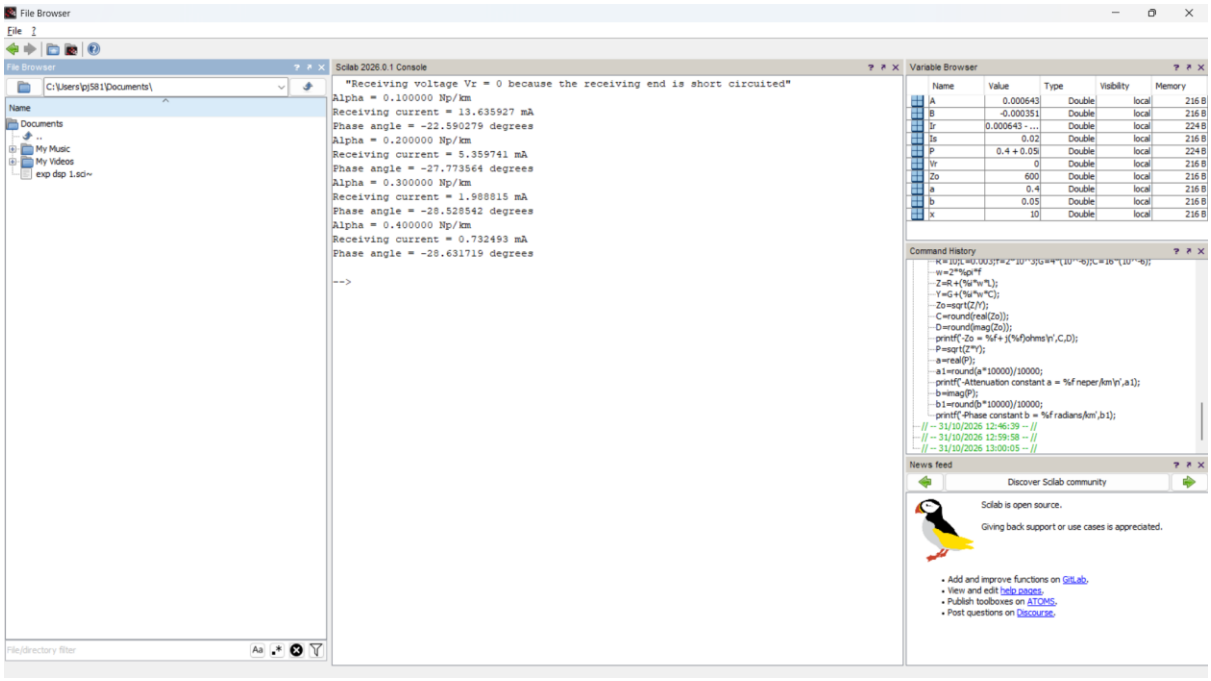


TEST CASE 2

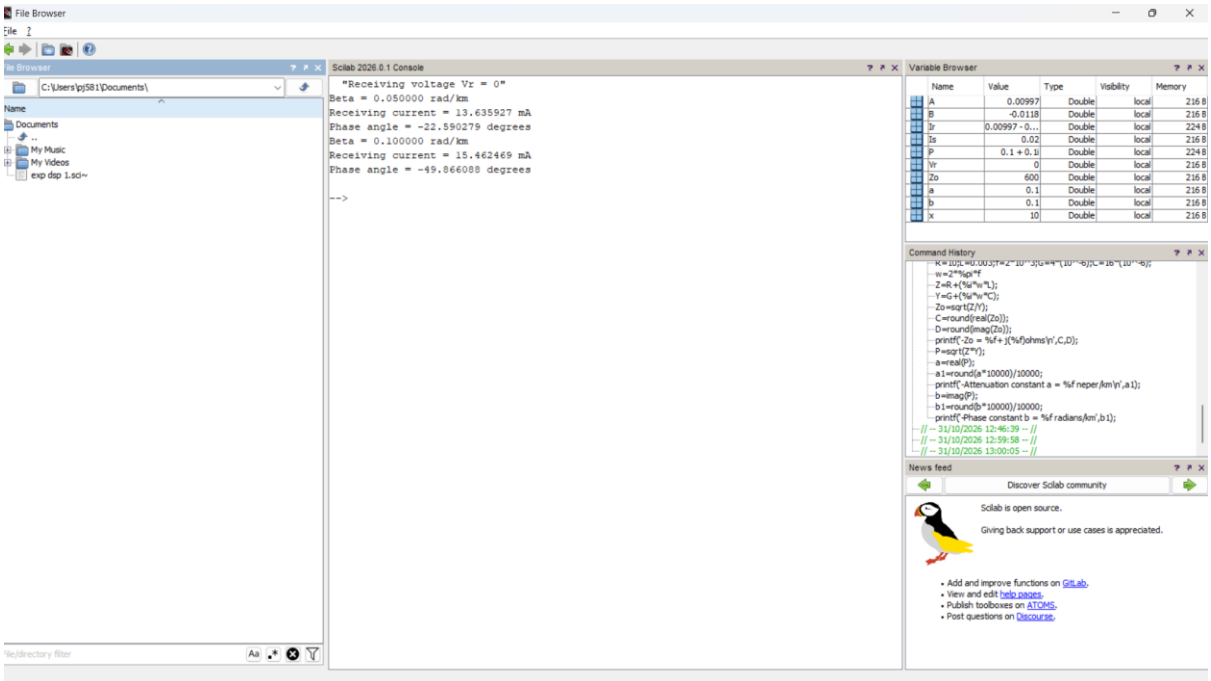


3. Evaluation of current in short circuited Lossless transmission line at the receiving end

TEST CASE 1

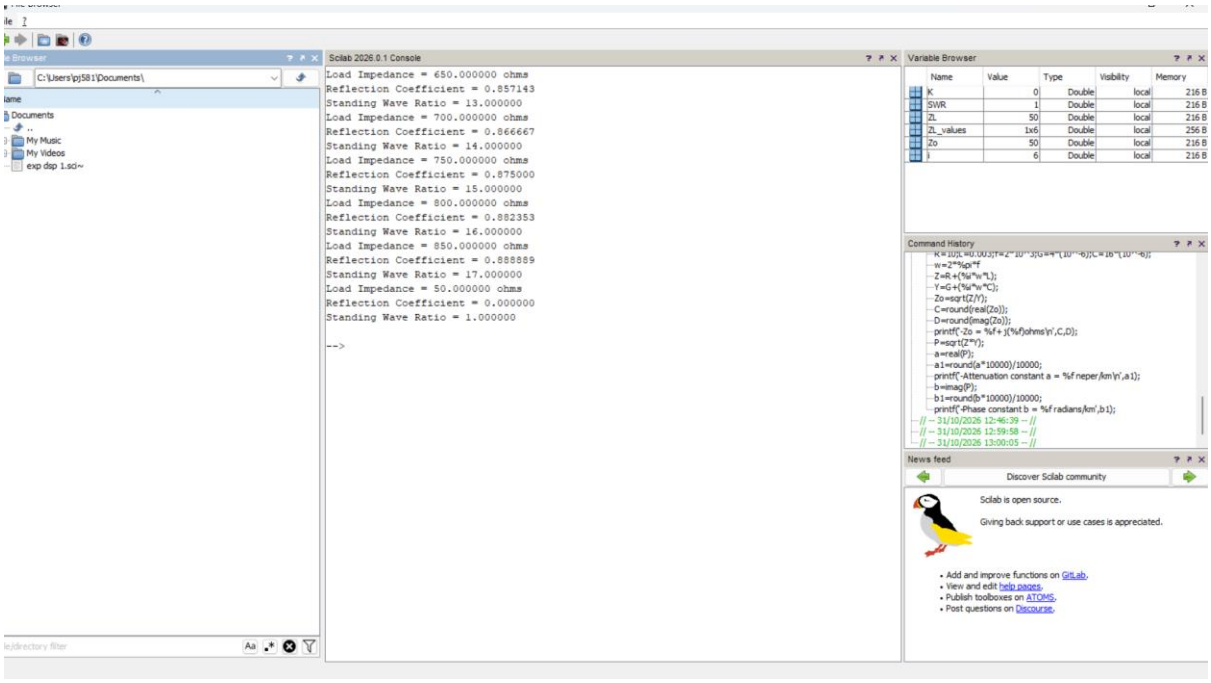


TEST CASE 2

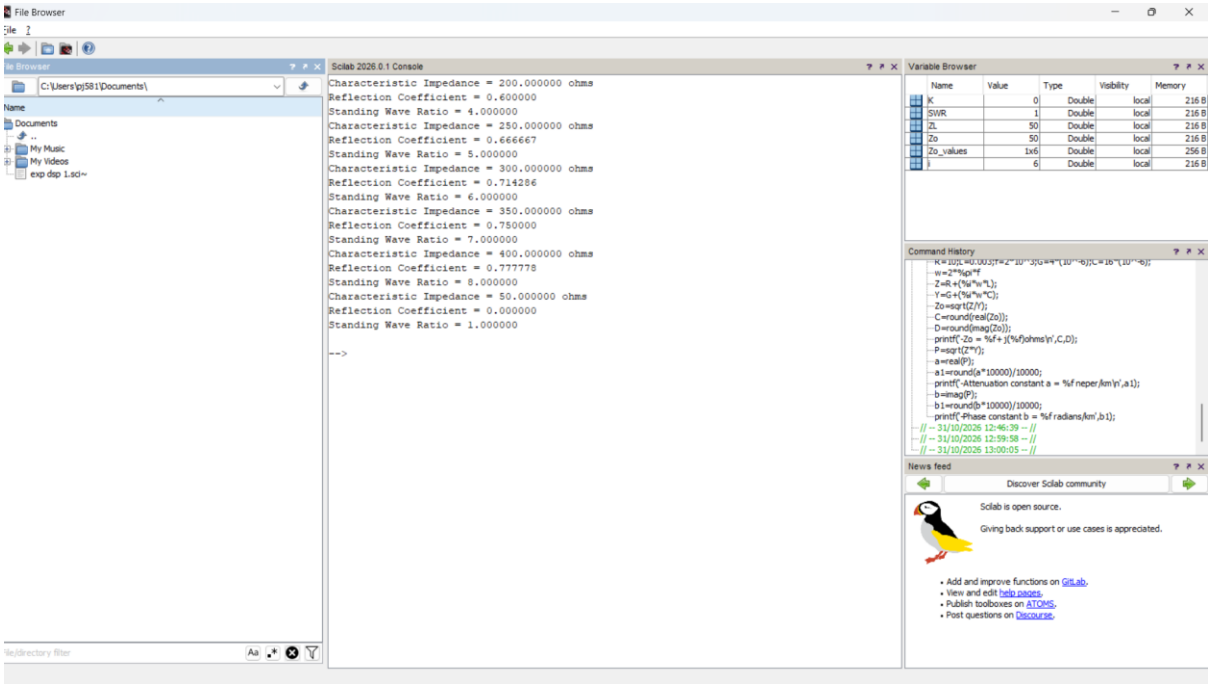


4. Analysis of SWR (Standing Wave Ratio)Transmission Lines by varying a)load impedance b) characteristic impedance

TEST CASE 1

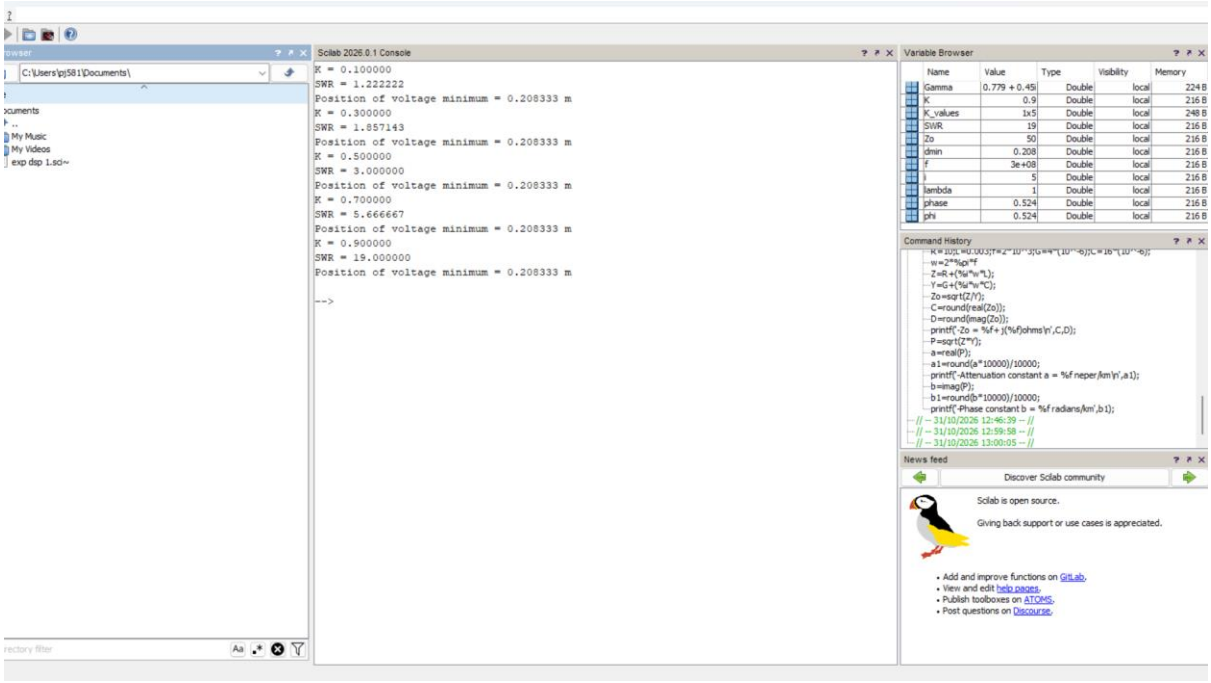


TEST CASE 2

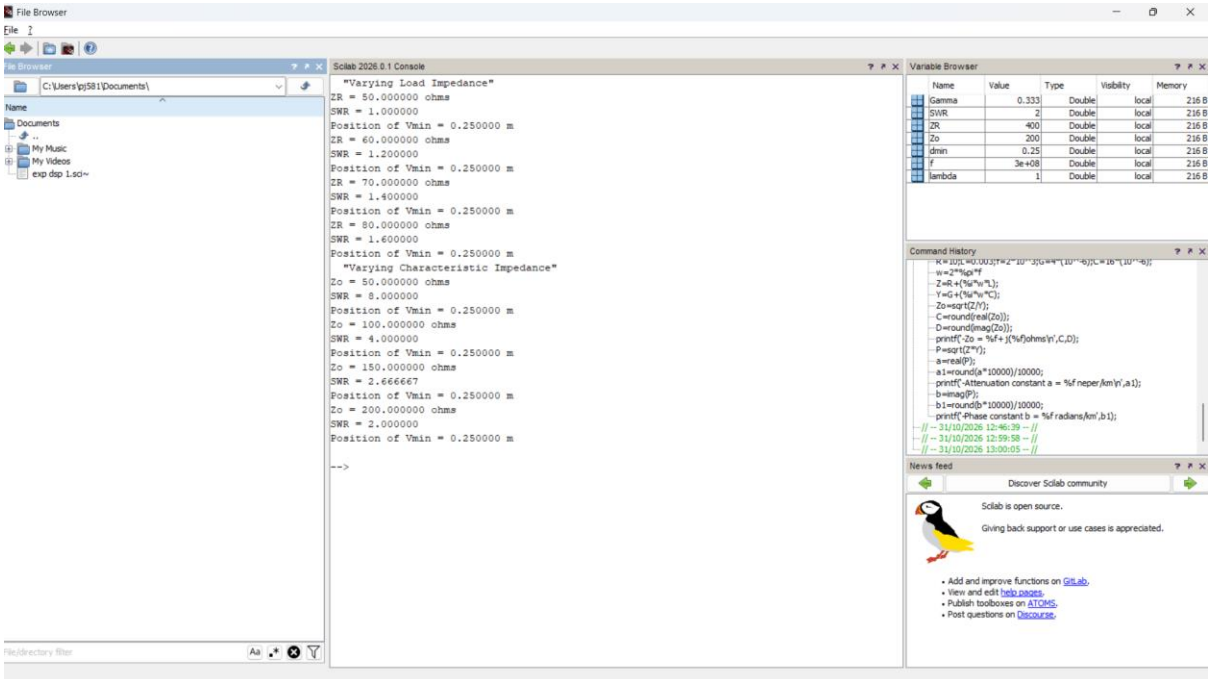


5. Exploration of the relationship between position of Voltage minima nearest to the load by varying magnitude of reflection coefficient K

TEEST CASE 1

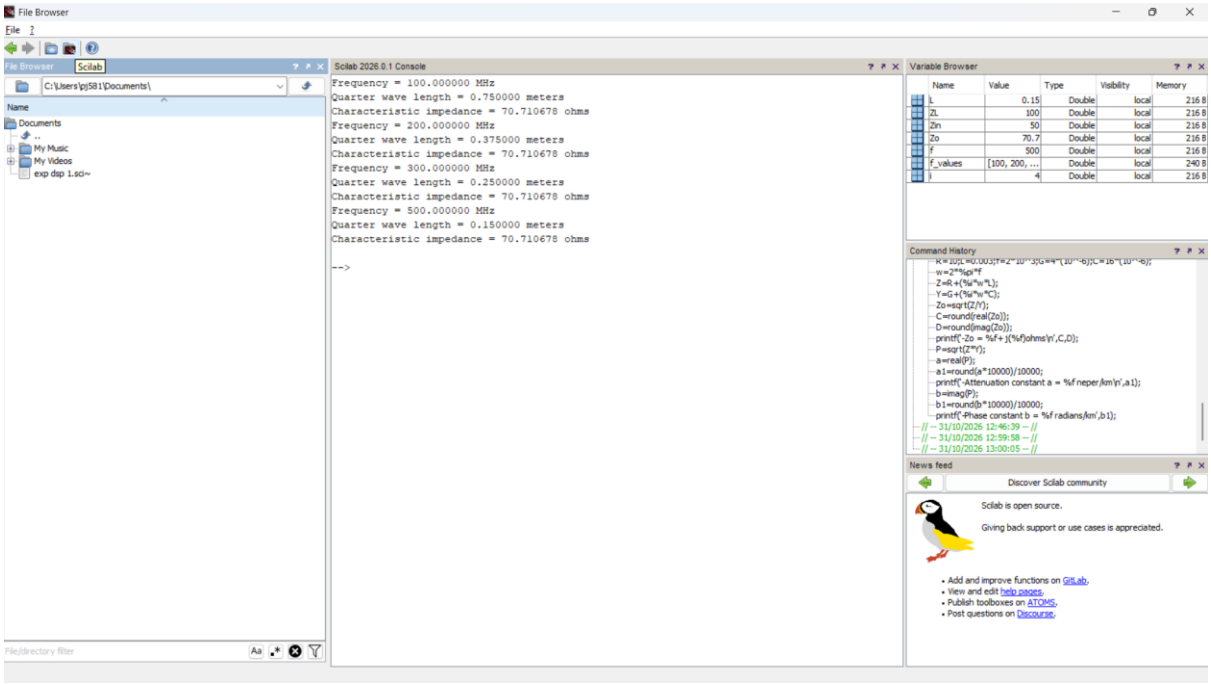


TEST CASE 2

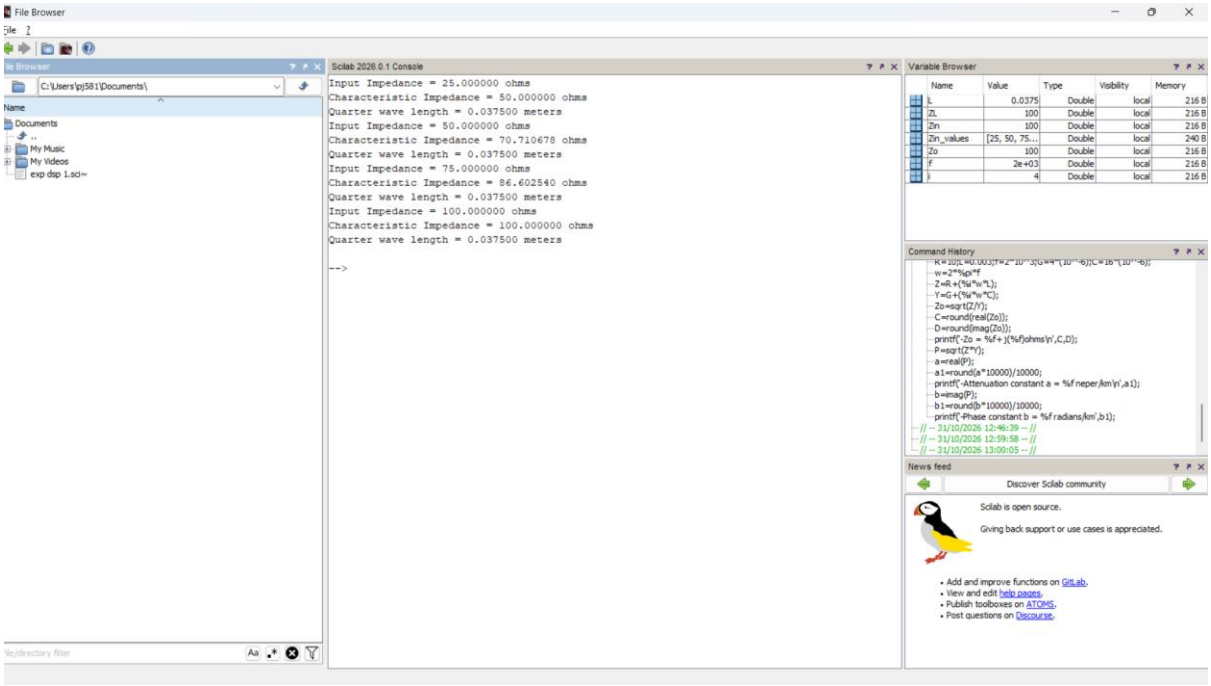


6. Designing a Quarter wave transformer by varying following parameters a) frequency b) input impedance c) characteristic impedance

TEST CASE 1

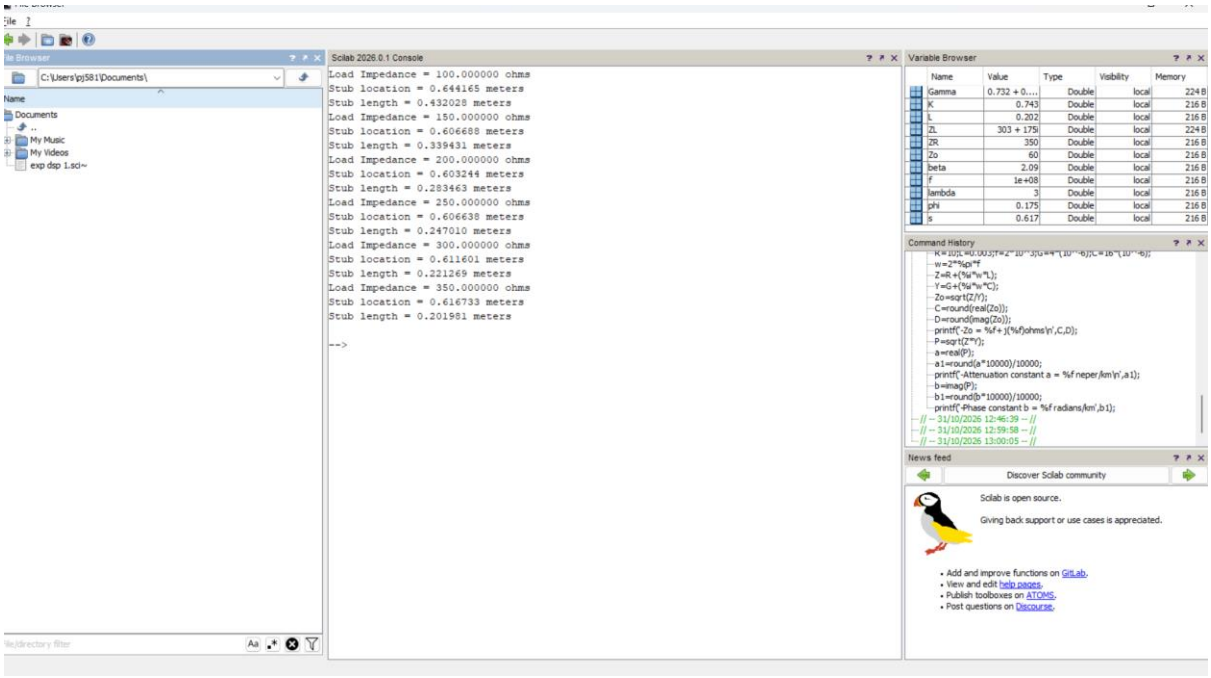


TEST CASE 2

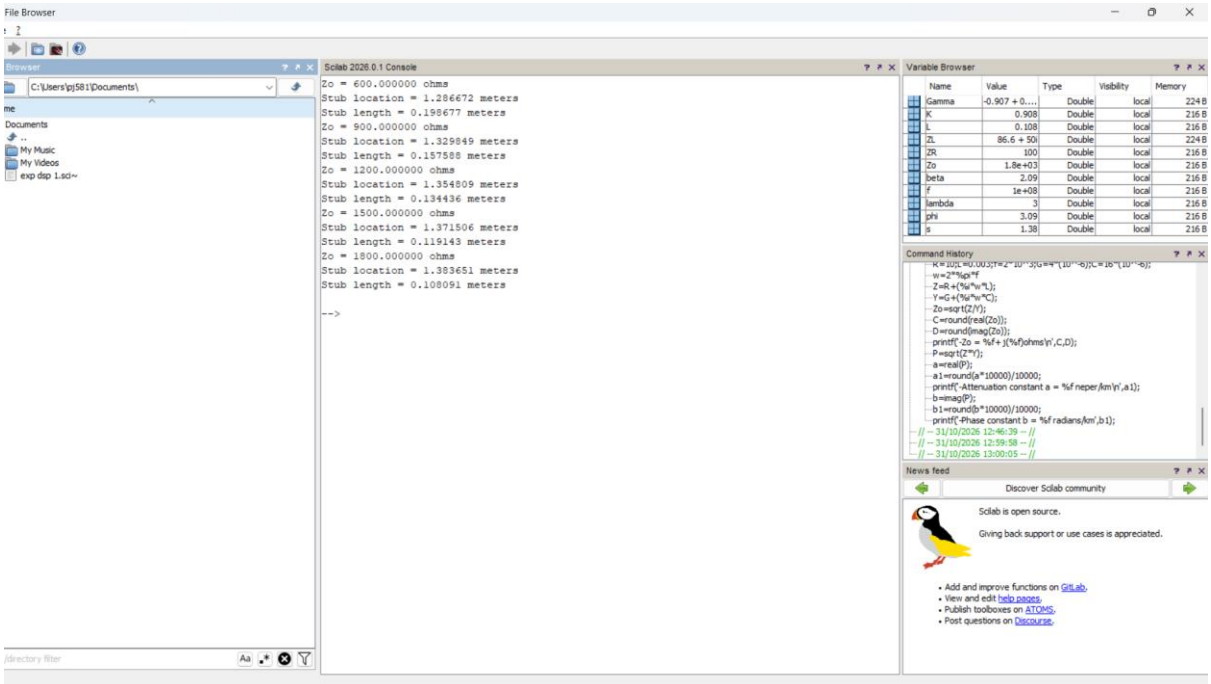


7. Strategize impedance matching with characteristic impedance and load impedance throughthe measurement of a) Location of the stub b) Length of the stu

TEST CASE 1

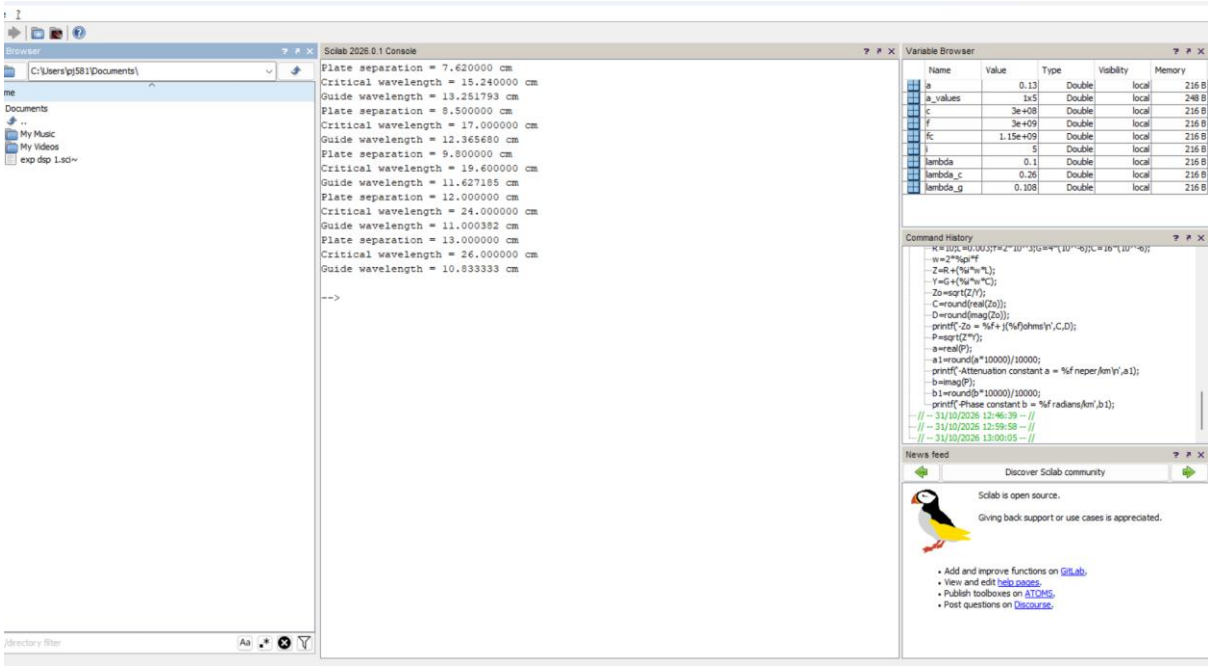


TEST CASE 2

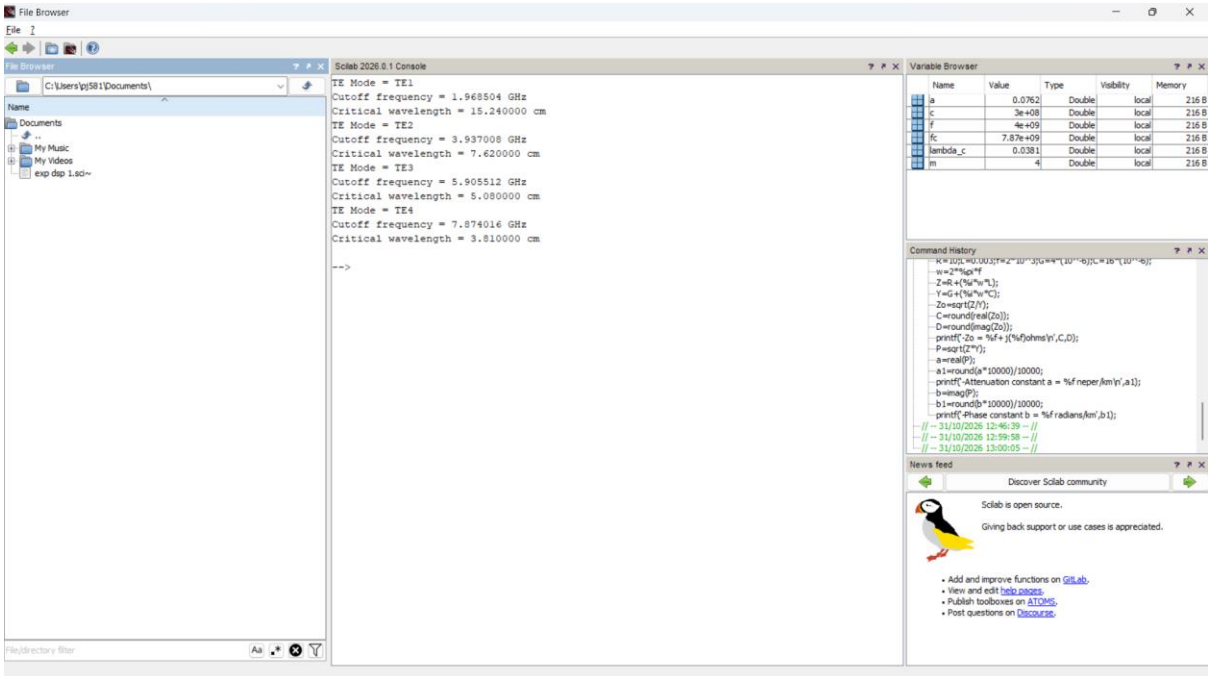


8. Analysis of Electromagnetic Waves across the two parallel plates estimating the following a) Critical Wavelength b) Guide Wavelength

TEST CASE 1

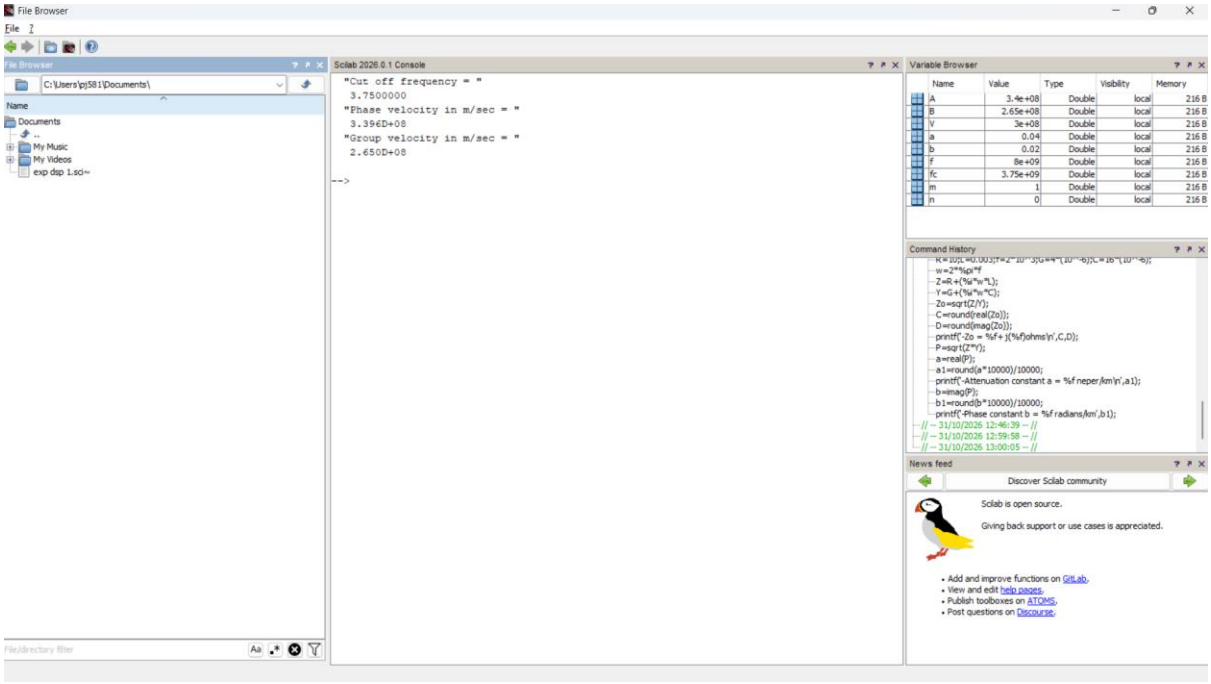


TEST CASE 2

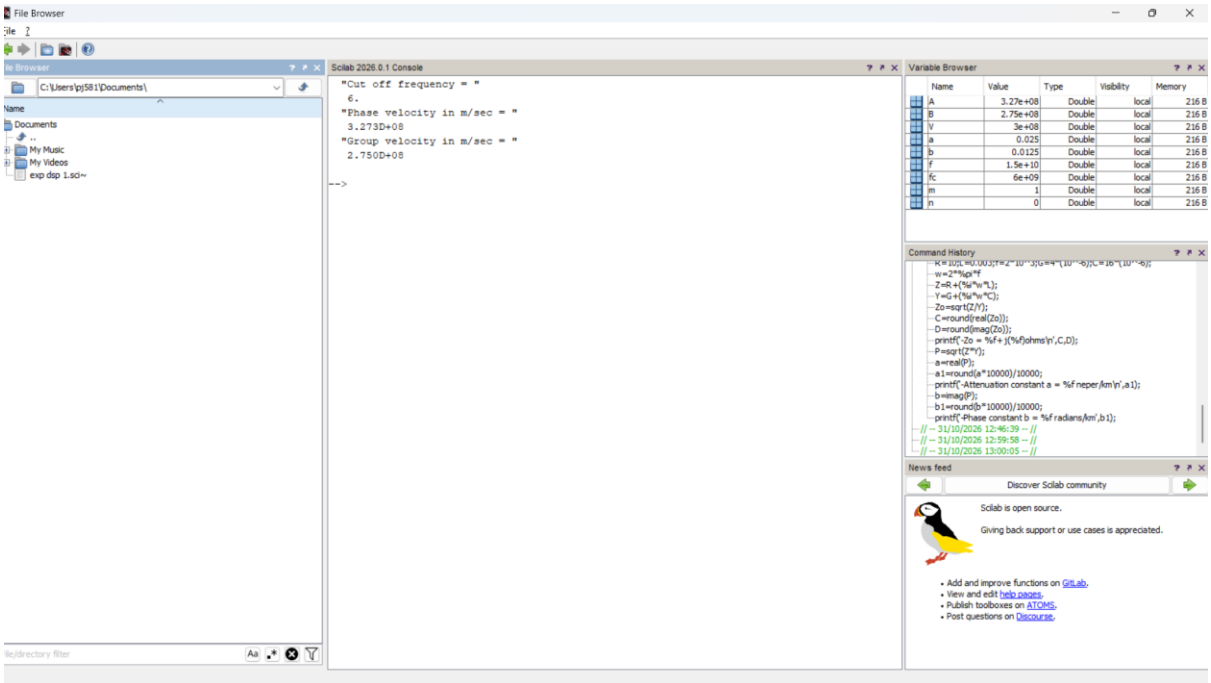


9. Calculation of Group velocity, Phase velocity and phase constant of EM waves in rectangular waveguides

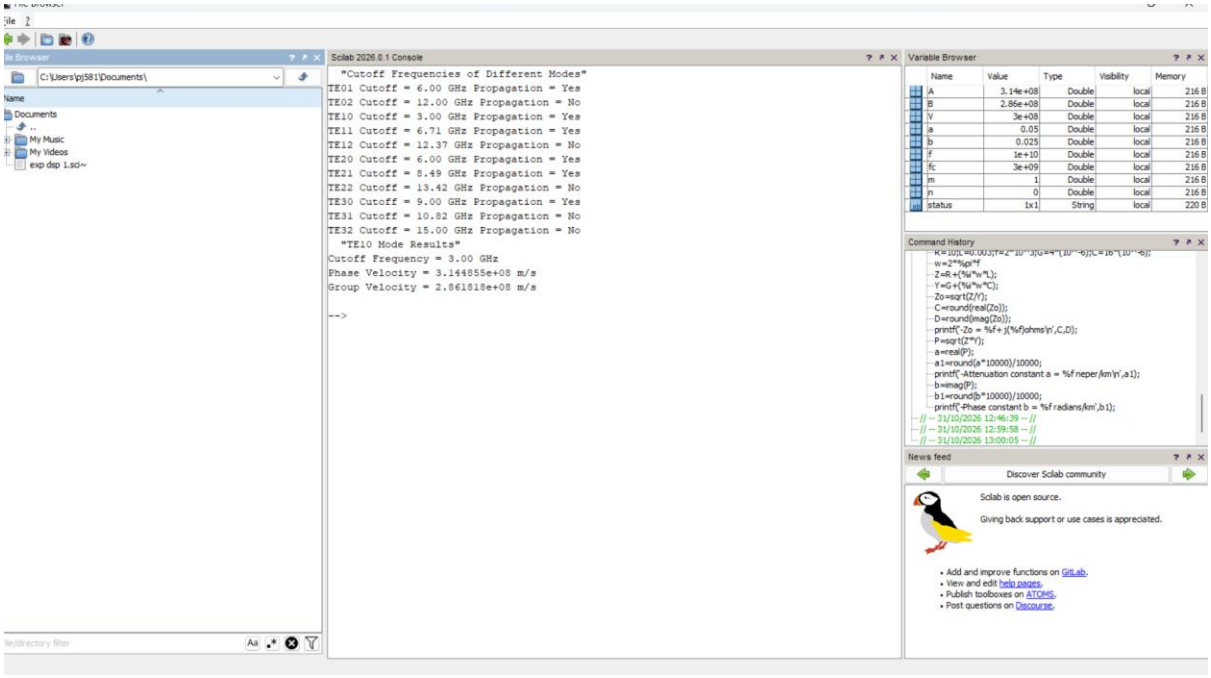
TEST CASE 1



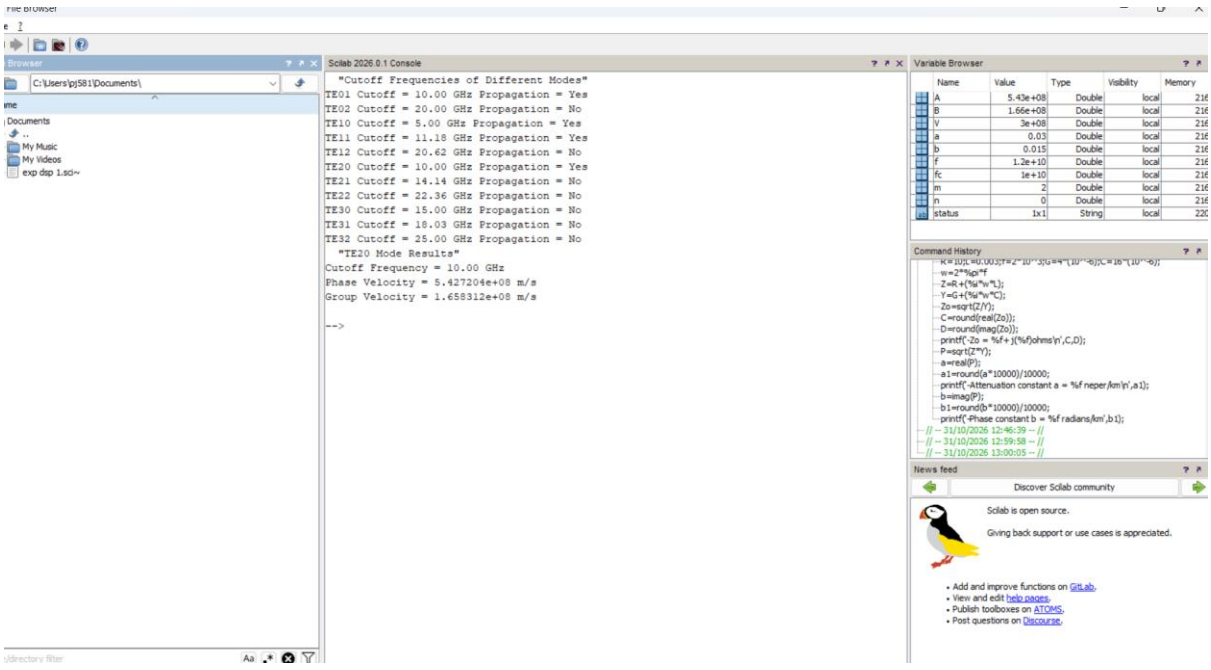
TEST CASE 2



TEST CASE 3

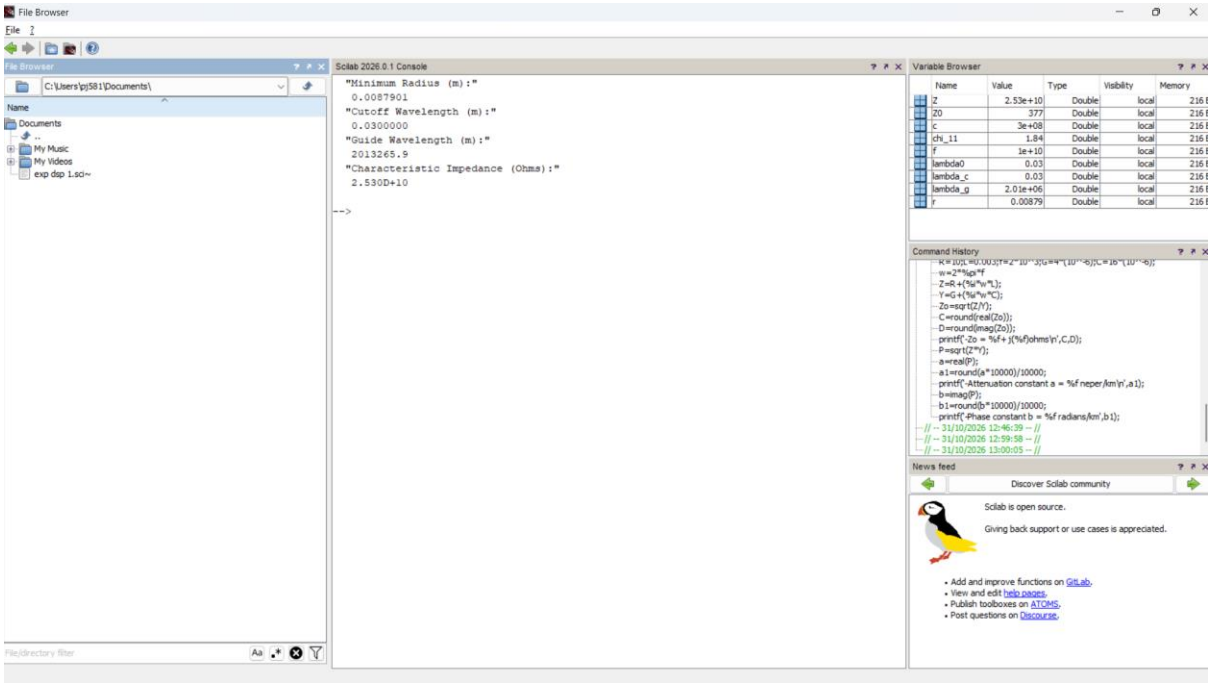


TEST CASE 4

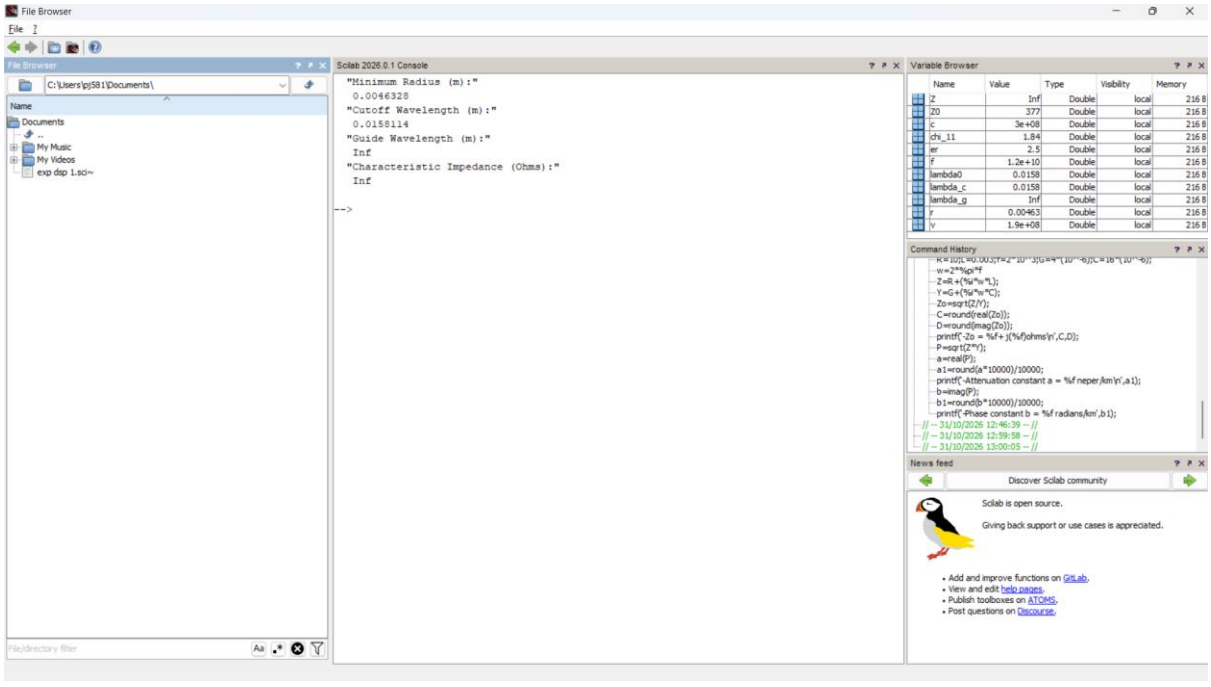


10. Examining the propagation characteristics of circular waveguide in terms of a) Cutoff wavelength d) Guide wavelength e) Characteristic impedance

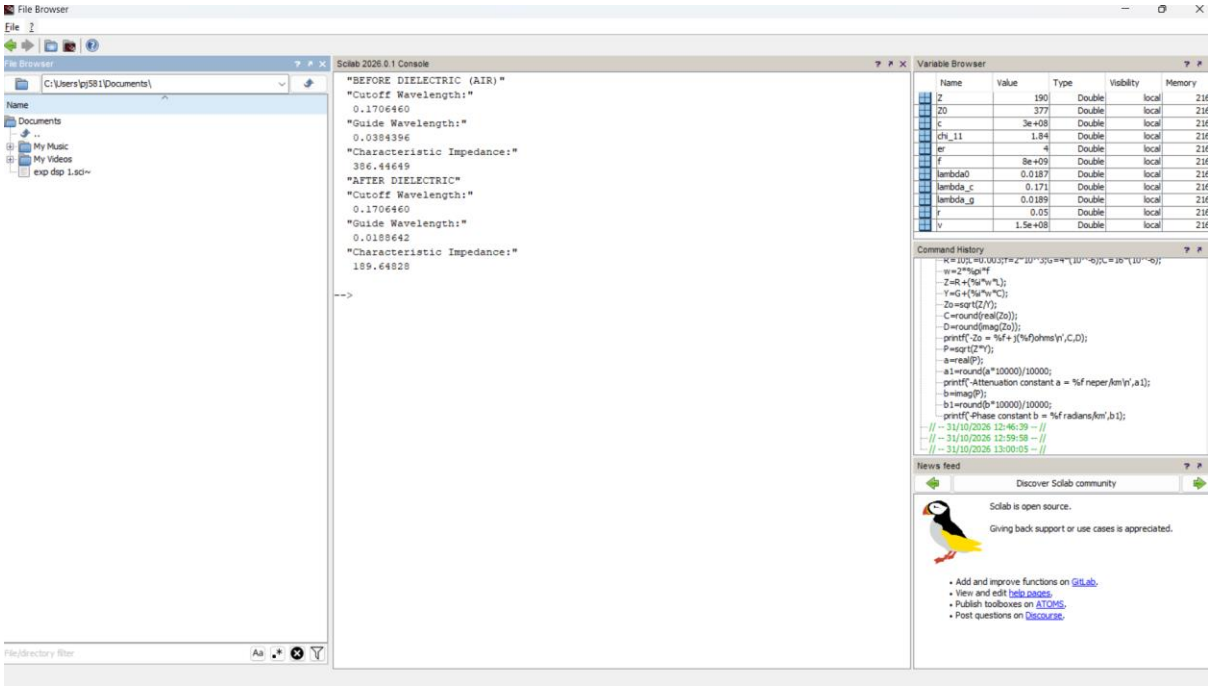
TEST CASE 1



TEST CASE 2



TEST CASE 3



TEST CASE 4

